Juan Armando Parra Flores

Guanajuato, México. juan.parra@cimat.mx juamnito

Education

M.Sc. **Mathematics.** CIMAT (Mathematics Research Center). Aug 2024 - July 2026 (expected)

- GPA (0-10 System): 10.0/10.0
- GPA translation: 4.0/4.0
- **Coursework:** Algebraic Topology, Fiber Bundle Geometry, Differential Topology, Functional Analysis.

B.Sc. **Mathematics.** University of Guanajuato. Aug 2019 – Dec 2023

- GPA (0-10 System) 9.73/10.0
- GPA translation: 3.89/4.0
- **Coursework:** Riemannian Geometry, Differential Geometry, Hyperbolic Geometry, Algebraic Topology, Algebraic Geometry, Measure Theory, Galois Theory, Statistical Methods, Algorithms, Introduction to AI, Computability Theory, Numerical Methods.

Research Experiences

1st Research School in Geometry at Cuernavaca. Workshops and courses in Heegard Splittings, Quandles, Algebraic Geometry and Homogeneous Spaces. Cuernavaca , Mex. June 2025

10th Oaxacan School of Mathematics. Geometric Group Theory team. Directed by Dr. Porfirio León, Dr. Nestor Colin and Dr. Rita Jiménez. Oaxaca, Mex. Jan - Feb 2025

- Handlebody Group and Heegaard Splittings.

Topologycal Aspects of the Simetries in Surfaces Directed by Dr. Israel Morales Jiménez Guanajuato, Mex Aug 2023 - Nov 2024

- CUBO University of La Frontera Chile (spanish) (paper).

MIT Summer Geometry Iniciative Worked in three projects: Online July - Aug 2024

- Poisson Algorithm of Surface Reconstruction. Directed by Dr. Silvia Sellán and Dr. Oded Stein.
- Voronoi piecewise surrogates. Directed by Dr. Martin Jorgensen.
- Neural Networks for Swept Volumes. Directed by Dr. Silvia Sellán and Dr. Noam Aigerman (project uploaded to SGI 2024 project reports).

Fields Undergraduate Summer Research Program. Predicting Performance of Machine Translation over Low Resource Languages. Directed by Dr. En-Shuin Annie Lee. Toronto, CA June - Aug 2023

- EACL 2024 (submission; arxiv).
- Transfered to this project, because of health reasons of the original advisor.

8th Oaxacan School of Mathematics. Topological Surfaces group. Directed by Dr. Israel Morales Jiménez. Oaxaca, Mex. Jan - Feb 2023

- Topological properties of the Mapping Class Group. (notes in spanish).

Skills

Programming Languages. Python, C++, C#, Bash, LaTeX.

Algorithms. Data structures and algorithms; Multilayer Perceptron; Optimization.

Software. Linux (Debian and Arch based distributions); git; makefile; vim.

Languages. Spanish (native); English (TOEFL iBT score 90/120).

Talks

Mexican Mathematical Society National Congress. The Handlebody Group and Heegaard Splittings. — BUAP Puebla, October 23, 2025

CIMAT Students Seminar. Consequences of the Alexander's Method. — CIMAT Guanajuato, March 21, 2025

10th Oaxacan School of Mathematics. Final project presentation. — UNAM Oaxaca, February 5, 2025

Jr. Seminar in the Mathematics Department of the University of Guanajuato. Surfaces, their homeomorphisms and a brief introduction to the Mapping Class Group. — UG Guanajuato, Nov 22, 2023

Seminar of Combinatorial Group Theory. Workshop in the Cayley Complexes (web page). — CIMAT Guanajuato, Nov 14, 2023

Fields Undergraduate Summer Research Program Presentation. Predicting Performance of Machine Translation over Low Resource Languages. — Fields Institute Toronto, Aug 9, 2023

8th Oaxacan School of Mathematics. Final project presentation. — UNAM Oaxaca, February 4, 2023

Teaching Assistantships

Reviewing and grading weekly homeworks; taking care of the grades; a weekly problem solving class; office hours one hour per week.

Differential and Integral Calculus III. Proof based differential calculus on several variables. — University of Guanajuato, Aug-Dec, 2025

Modeling with Differential Geometry. Differential Geometry of Curves and Surfaces with applications. — CIMAT, Aug-Dec, 2024

Differential and Integral Calculus III. Proof based differential calculus on several variables. — University of Guanajuato, Aug-Dec, 2023

Topology. Point set topology. — University of Guanajuato, Aug-Dec, 2022

Differential and Integral Calculus II. Proof based integral calculus on a single variable. — University of Guanajuato, Jan-July, 2022

Differential and Integral Calculus III. Proof based differential calculus on several variables. — University of Guanajuato, Aug-Dec, 2021

Tutoring

Tutor in Calculus Problem Solving Workshop. Tutoring a group of first-semester undergraduate students from different mathematics related programs in Mexico. We lead them during a week to solve hard calculus problems and present their work. — CIMAT, July, 2024

Independent Math Teacher. — Guanajuato, Jan-Aug 2024

Tutor in Math Writting Workshop. Assisting first-semester students to understand and reproduce proofs in mathematics. — University of Guanajuato, Aug-Sep, 2020

Service

Alumni Society of the University of Guanajuato Mathematics Department. President.	University of Guanajuato May 2023-Apr 2024
Campus Guanajuato Council Member of the University of Guanajuato. Substitute.	University of Guanajuato Feb 2021-Feb 2023
Homeworks workshop tutor for elementary school students. At the *Buen Pastor*'s Convent.	Guanajuato Jan-July 2022
Coach in the Guanajuato Mathematical Olympiad. For elementary and middle school levels.	Guanajuato Jan-July 2021
Matemorfosis Team at CIMAT. CIMAT's divulgation of math and science team.	CIMAT Aug 2019-July 2020

Scholarships

Master's studies scholarship. Given by the CONAHCYT to students enrolled in a science graduate program.	CONAHCYT Aug 2024 - July 2026
Excellence Academic Scholarship. Awarded by a joint agreement of CIMAT with the University of Guanajuato.	CIMAT-UG Aug 2019 - Dec 2023
Highschool Merit Acknowledge Scholarship.	University of Guanajuato Aug 2018 - July 2019

Honors and awards

Trajectory Merit Acknowledge at University of Guanajuato. Awarded by the University of Guanajuato to students with the highest grades during their whole trajectory.	University of Guanajuato 2024
Annual Merit Acknowledge by the University of Guanajuato. Awarded by the University of Guanajuato to students with the highest grades of their program during their academic year.	University of Guanajuato 2020, 2022 and 2023
Highschool Merit Acknowledge by the University of Guanajuato. Awarded by the University of Guanajuato to students with the highest grades of their program during their academic year.	University of Guanajuato 2019

Attendance to conferences and courses

School "Fico González Acuña" of knots and 3-manifolds.	UNAM Oaxaca Dec 2025
VIII Conference on Finite Dimensional Integrable Systems in Geometry and Mathematical Physics (FDIS).	CIMAT Guanajuato Aug 2025
1st Meeting in Geometry, Algebra and Topology.	UNAM Oaxaca June 2025
School "Fico González Acuña" of knots and 3-manifolds.	UNAM Cuernavaca Dec 2024
EMALCA (School of Mathematics of Latin America and the Caribbean) (web page).	UNAM Cuernavaca June 2024
CIMAT Thematic School. Topology and Differential Geometry.	CIMAT Guanajuato April 2024
CIMAT Thematic School. Algebra and Algebraic Geometry.	CIMAT Guanajuato Dec 2023

Little Conference of Differential Geometry. CIMAT Guanajuato.	CIMAT Guanajuato Dec 2023
CIMAT Thematic School. Topology and Differential Geometry.	CIMAT Guanajuato June 2023
School "Fico González Acuña" of knots and 3-manifolds.	CIMAT Guanajuato Dec 2022
Mathematical Semesters at CIMAT. Data Science Summer Course.	CIMAT Guanajuato July 2022